



Project Proposal: Blood Donation & Transfusion Database System

Course: Fundamentals of Database Systems (CSE462a / CSE242)

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Problem Statement

The process of managing blood donations, tracking inventory, and fulfilling hospital transfusion requests is highly time-sensitive and critical to patient survival. Manual or fragmented tracking systems are likely to include human error, which can lead to life-threatening delays or incompatible blood transfusions. There is a crucial need for a centralized, highly optimized relational database system capable of securely managing donor records, maintaining real-time blood type inventory, and automating the complex logic required to instantly match available blood units with compatible patients in emergency scenarios.

Objectives

- **Design a Normalized Relational Schema:** Structure the database to eliminate data redundancy and ensure strict data integrity across donor, inventory, and hospital records.
- **Ensuring Data Consistency:** Utilize robust foreign key constraints to maintain accurate relationships between donors, their specific blood units, and the hospitals receiving them.
- **Automate Compatibility Logic:** Develop secure stored procedures that automatically handle the complex medical rules for blood type compatibility, ensuring patients only match with safe donors.
- **Optimize Query Performance:** Implement indexing and query optimization techniques to guarantee rapid retrieval of available blood units during urgent, time-critical hospital requests.

Expected Features

- **Comprehensive Donor Management:** Secure records of donor profiles, contact information, eligibility statuses, and historical donation dates.
- **Real-Time Blood Type Inventory:** A dynamic tracking system that monitors the exact quantity, blood type, and expiration date of all blood units currently held in the bank.
- **Hospital Request Pipeline:** A feature allowing registered hospitals to submit, track, and update the status of specific blood transfusion requests.
- **Transfusion History & Tracking:** An immutable log of all completed transactions, tracing every blood unit from the specific donor to the receiving patient.
- **Automated Matching System:** Stored procedures that instantly cross-reference hospital requests with the current inventory, filtering for strict ABO/Rh blood type compatibility and oldest-in, first-out (OIFO) expiration logic.